

Joren Gijsbrechts

+34 696 949 771 | joren.gijsbrechts@esade.edu | [LinkedIn](#) | [Google Scholar](#) | [GitHub](#) | [Website](#)

SUMMARY

As an Associate Professor of Operations Management at Esade Business School, I create algorithms that translate complex datasets into actionable decisions. I also deliver custom talks on AI, analytics, and Supply Chain 4.0.

EDUCATION

University of Antwerp <i>B.Sc. & M.Sc. Applied Economics</i>	Antwerp, Belgium <i>Sep. 2009 – June 2012</i>
KU Leuven <i>Ph.D. Operations Management</i>	Leuven, Belgium <i>Sep. 2016 – August 2020</i>
Kellogg School of Management <i>Visiting PhD student</i>	Evanston, Illinois <i>Sep. 2019 – Dec. 2019</i>

ACADEMIC EXPERIENCE

Assistant Professor <i>Católica Lisbon School of Business and Economics</i>	Sep. 2020 – Aug. 2023 <i>Lisbon, Portugal</i>
Visiting Scholar <i>MIT Sloan School of Management</i>	Jan. 2023 – March. 2023 <i>Cambridge, Massachusetts</i>
Assistant Professor <i>Esade Business School, Ramon Llull University</i>	Sep. 2023 – Aug. 2025 <i>Barcelona, Spain</i>
Associate Professor <i>Esade Business School, Ramon Llull University</i>	Sep. 2025 – Present <i>Barcelona, Spain</i>

INDUSTRY EXPERIENCE

Supply Chain Analyst <i>Procter and Gamble</i>	June. 2014 – Aug. 2016 <i>Stockholm, Sweden</i>
Advisor, Algorithms <i>CurveCatch</i>	March 2020 – Present <i>Antwerp, Belgium</i>

PUBLICATIONS

Peer-Reviewed Publications

- Cohen, M. C., Dai, T., Perakis, G., Agrawal, N., Allon, G., Boute, R. N., Gijsbrechts J., ..., Zhang, D.. (2026) Supply Chain Management in the AI Era: A Vision Statement from the Operations Management Community. *Manufacturing Services & Operations Management (Forthcoming)* .
- De Moor B, Boute RN, Gijsbrechts J. Discount replenishment opportunities with limited capacity: Optimal policy and pricing dynamics. *European Journal of Operational Research (Forthcoming)*
- Farias V, Gijsbrechts J, Khojandi AI, Peng T, Zheng A. Speeding up Policy Simulation in Supply Chain RL. *Proceedings of the 42nd International Conference on Machine Learning, in Proceedings of Machine Learning Research* 267:16161–16177 (2025). Available from <https://proceedings.mlr.press/v267/farias25a.html>
- Gijsbrechts, J., Boute, RN, Disney, SM, Van Mieghem, JA (2025). Volume Flexibility at Responsive Suppliers in Reshoring Decisions: Analysis of a Dual Sourcing Inventory Model. *Production and Operations Management*, **34**(5), 1114–1133. DOI: 10.1111/poms.13719
- Gijsbrechts J, Van Staden HE. Where Did the Demand Go? Teaching Demand Censoring with the Newsvendor Challenge. *INFORMS Transactions on Education* (2024). DOI:10.1287/ited.2024.0085

- **Gijsbrechts J, Imdahl C, Boute RN, Van Mieghem J.** Optimal Robust Inventory Management with Volume Flexibility: Matching Capacity and Demand with the Lookahead Peak-Shaving Policy. *Production and Operations Management* 32(11):3357–3373 (2023). DOI:10.1111/poms.14069
- **Gijsbrechts J, Boute RN, Van Mieghem JA, Zhang DJ.** Can Deep Reinforcement Learning Improve Inventory Management? Performance on Dual Sourcing, Lost Sales and Multi-Echelon Problems. *Manufacturing & Services Operations Management* 24(3):1261–1885 (2022). DOI:10.1287/msom.2021.1064
- **Boute RN, Disney SM, Gijsbrechts J, Van Mieghem JA.** Dual Sourcing and Smoothing under Non-Stationary Demand Time Series: Re-shoring with SpeedFactories. *Management Science* 68(2):1039–1057 (2021). DOI:10.1287/mnsc.2020.3951
- **De Moor B, Boute RN, Gijsbrechts J.** A reward shaping approach to improve the applicability of deep reinforcement learning to inventory management. *European Journal of Operational Research* 301(2):535–545 (2021). DOI:10.1016/j.ejor.2021.10.045
- **Boute RN, Gijsbrechts J, van Jaarsveld W, Vanvuchelen N.** Deep reinforcement learning for inventory control: A roadmap. *European Journal of Operational Research* 298(2):401–412. DOI:10.1016/j.ejor.2021.07.016
- **Yee H, Boute RN, Gijsbrechts J.** Synchromodal Planning under Stochastic Travel Times. *Computers in Industry* 125:103367 (2021). DOI:10.1016/j.compind.2020.103367
- **Vanvuchelen N, Gijsbrechts J, Boute RN.** Use of Proximal Policy Optimization for the Joint Replenishment Problem. *Computers in Industry* 119:103239 (2020). DOI:10.1016/j.compind.2020.103239

Working Papers

- **Farias V, Gijsbrechts J, Khojandi A, Peng T, Zheng A.** Speeding up Policy Simulation in Supply Chain RL. arXiv:2406.01939
- **Helm PP, Doerr JN, Gijsbrechts J, Minner S.** Hard Constraints, Smooth Gradients: Learning Feasible Policies in General Inventory Networks via Differentiable Projection.
- **Bertani N, Gijsbrechts J, Godinho de Matos M, Wagner L.** Every Man is an Island: Reminder Timing for Heterogeneous Mobile Users. *Available at SSRN* (2024). DOI: 10.2139/ssrn.4911622
- **Gijsbrechts J, Olivares Nadal A, Wagner L.** Replenishing a Try-Before-You-Buy Channel in Online Retail.
- **Gijsbrechts J, van Jaarsveld W.** The Efficiency Paradox in Agentic AI Workflows: When Optimization Increases the Electricity Bill. *Available at SSRN* (2026). DOI: 10.2139/ssrn.6168735
- **Wagner L, Godinho de Matos M, Gijsbrechts J, Amorim P.** When the Online Store Goes Dark: Customer Responses to a Cyberattack in Grocery Retail. *Available at SSRN* (2026). DOI: 10.2139/ssrn.6653138

Book Chapters

- **Boute RN, Gijsbrechts J, Van Mieghem JA.** Digital Lean Operations: Smart Automation and Artificial Intelligence in Financial Services. In: Babich V, Birge J, Hilary G (eds) *Innovative Technology at the interface of Finance and Operations*. Springer Series in Supply Chain Management, Springer Nature (2021).
- **Gijsbrechts J, Boute RN, Van Mieghem JA, Zhang D.** AI in Inventory Management: The Disruptive Era of DRL and Beyond. Forthcoming chapter in “AI in Supply Chains: Perspectives from Global Thought Leaders,” edited by Maxime C. Cohen and Tinglong Dai, and part of the Springer Series in Supply Chain Management (edited by Prof. Chris Tang).

Current Teaching

Business Analytics <i>Esade Business School – Full-time MBA</i>	2023–present
Supply Chain Management <i>Esade Business School – Full-time MBA</i>	2023–present
Data Science and Operations for Sustainability <i>Esade Business School – MSc Sustainability</i>	2023–present
Simulation, Optimization and Computational Methods <i>Esade Business School – Master of Research</i>	2025–present
Business Analytics <i>Esade Business School – Executive-MBA</i>	2025–present

Invited Teaching

Supply Chain 4.0: AI, Data & Smart Decision-Making <i>Antwerp Management School, Antwerp, Belgium – Executive Masterclass</i>	2025–present
Data-driven Decision-Making and Business Analytics <i>CLSBE, Lisbon, Portugal – Executive-MBA</i>	2022–present

Past Teaching

Data-Driven Organizations <i>Esade Business School – Custom Program for GB Foods</i>	2025
Green Transport and Machine Learning <i>KU Leuven Kortrijk, Belgium – Postgraduate Smart Logistics</i>	2019–2024
Inventory Control and Machine Learning in Supply Chains <i>University of Antwerp, Belgium – Executive Masters Program</i>	2019–2023
Operations Management <i>CLSBE, Lisbon, Portugal – Executive Masters Program</i>	2021–2022
Supply Chain Management <i>CLSBE, Lisbon, Portugal – M.Sc. Business Analytics</i>	2021–2023
Advanced Predictive Analytics <i>CLSBE, Lisbon, Portugal – M.Sc. Business Analytics</i>	2022–2023
Operations Analytics <i>CLSBE, Lisbon, Portugal – M.Sc. Business Analytics</i>	2022–2023
Introduction to Operations Management <i>CLSBE, Lisbon, Portugal – B.Sc. Business Administration and Applied Economics</i>	2020–2023
Data-driven Decision Making and Business Analytics <i>CLSBE, Lisbon, Portugal – B.Sc. Business Administration and Applied Economics</i>	2022–2023
Deep Reinforcement Learning in Inventory Management <i>NYU Stern School of Business, US – PhD Program</i>	2022
Deep Reinforcement Learning in Inventory Management <i>Wharton, US – PhD Program</i>	2022
Impact of Machine Learning on Operations Management <i>Vlerick Business School, Belgium – Supply Chain Bootcamp</i>	2019
ML in Operations Management <i>Catholic University Milan, Italy – Workshop</i>	2019

SERVICE

Reviewer for *Management Science*, *Operations Research*, *Manufacturing Services & Operations Management*, *Production and Operations Management*, *European Journal of Operational Research*, *International Journal of Production Economics*, *Omega*, *INFORMS Journal on Computing*.